

Practical Pipeline Issues

Access the github code repository

- Current repository is private and you need a github account to gain access
 - I will make it public, but didn't want to do so quite yet.
- Clone the repository to your working directory using git pull requests.
 - I haven't yet distinguished stable versions of the repository.
 - However, the current version is the correct one to clone.

Create the MNE-Python environment

- I have a 'requirements.txt' file for mne-python on the HPC and MBPro
- I last built my mne environment using 'pip' since 'conda' didn't yet have the latest version of mne-python. Here are the commands:
 - `conda create -n mne python=3.11`
 - `conda activate mne`
 - `pip install --upgrade pip setuptools wheel`
 - `pip install numpy scipy matplotlib pandas h5py scikit-learn threadpoolctl`
 - `pip install mne mne-bids autoreject mne-qt-browser`
 - `pip install pyvista pyvistaqt vtk pyqtgraph`
 - `pip install PyQt6`
 - `pip install ruamel.yaml`

Make sure you have an account on HPC Milgram

- You must use Yale's VPN to access Milgram. This is true even if you are on the Yale network.
 - Milgram does not respond to 'ping' unless you are on the VPN.
- Milgram requires quarterly access audits and current HIPAA authorization.
 - If these are not done in a timely way by your PI, you lose full access without any warning.

There are two ways to access Milgram:

- Open OnDemand (**OOD**) provides a graphical user interface (GUI) to Milgram that runs in a browser
 - `ood.milgram.ycrc.yale.edu`

- The GUI is serviceable, but not sophisticated.
- You can run interactive jobs on OOD, but interactive graphics are slow.
- **SSH** (secure shell) provides a command line interface (CLI) to Milgram.
 - You must set up an encrypted key for each computer you use to connect via SSH.
 - SSH access is necessary for the pipeline's **hybrid mode**.
 - SSH access will prompt for DUO two-factor authentication.
 - This is also true when accessing the HPC with scp or rsync.

The pipeline assumes your data is in a BIDS directory structure.

- Presently, our data are not organized in BIDS format
- We need to enforce this for data provenance reasons.

BIDS MEG Directory Structure

```
BIDS_Dataset/
|-- dataset_description.json
|-- participants.tsv
|-- participants.json
|-- README
|-- CHANGES
|-- .bidsignore
|-- code/
|   |-- test_bids_validation.py
|-- derivatives/
|   |-- preprocessing/
|   |   |-- dataset_description.json
|   |   |-- sub-01/
|   |   |   |-- ses-001/
|   |   |   |   |-- meg/
|   |   |   |       |-- sub-01_ses-001_task-rest_desc-preproc_meg.fif
|   |   |   |       |-- sub-01_ses-001_task-rest_desc-preproc_meg.json
|   |   |   |       |-- sub-01_ses-001_task-rest_desc-ica_components.tsv
|   |   |-- sub-02/
|-- sub-01/
|   |-- sub-01_sessions.tsv
|   |-- ses-001/
|   |   |-- meg/
|   |   |   |-- sub-01_ses-001_task-rest_meg.fif
|   |   |   |-- sub-01_ses-001_task-rest_meg.json
```

```

|   |   |   |-- sub-01_ses-001_task-rest_channels.tsv
|   |   |   |-- sub-01_ses-001_task-rest_events.tsv
|   |   |   |-- sub-01_ses-001_coordsystem.json
|   |   |   |-- sub-01_ses-001_headshape.pos
|   |   |   |-- sub-01_ses-001_photo.jpg
|   |   |   |-- sub-01_ses-001_task-visualoddball_meg.fif
|   |   |   |-- sub-01_ses-001_task-visualoddball_meg.json
|   |   |   |-- sub-01_ses-001_task-visualoddball_channels.tsv
|   |   |   |-- sub-01_ses-001_task-visualoddball_events.tsv
|   |   |   |-- sub-01_task-mmn_run-01_meg.fif
|   |   |   |-- sub-01_task-faces_acq-crosstalk_meg.fif
|   |   |-- anat/
|   |   |   |-- sub-01_ses-001_T1w.nii.gz
|   |   |   |-- sub-01_ses-001_T1w.json
|   |-- ses-002/
|       |-- meg/
|-- sub-02/
|   |-- [similar structure to sub-01]
|-- sub-emptyroom/
    |-- ses-001/
        |-- meg/
            |-- sub-emptyroom_ses-001_task-noise_meg.fif
            |-- sub-emptyroom_ses-001_task-noise_meg.json

```

Required BIDS Files

dataset_description.json - Dataset metadata and BIDS version

participants.tsv - Subject demographics and information

meg recording files - Raw MEG data in vendor format (.fif, .ds, .4d)

meg.json sidecars - Recording parameters and metadata

channels.tsv - Channel locations, types, and status

Optional BIDS Files

events.tsv - Stimulus presentation and behavioral events

coordsystem.json - Coordinate system definitions and transformations

headshape.pos - Digitized head shape points

photo.jpg - Head position reference photos

sessions.tsv - Session-level information

Common Test Files in MEG Datasets

sub-emptyroom recordings - Background noise measurements

task-mmN paradigms - Mismatch negativity protocols

acq-crosstalk data - Cross-talk compensation measurements

multiple run files - Repeated task acquisitions

MEG Vendor File Formats

Elekta/Neuromag - .fif files

CTF - .ds directories with multiple internal files

BTi/4D - .pdf header + .4d data files

KIT/Yokogawa - .sqd/.con files

Key BIDS Naming Conventions

sub-XX - Subject identifier

ses-XX - Session identifier

task-XX - Task or paradigm name

run-XX - Run number for repeated acquisitions

acq-XX - Acquisition parameters or type

desc-XX - Processing description for derivatives